

Ramesh's Homelab & Media Server Documentation

Overview

This document describes the complete homelab environment maintained by Ramesh.

The environment is built around a Raspberry Pi 3B running CasaOS and serves multiple purposes:

- Media automation
 - Media streaming
 - Network-wide ad blocking
 - Service monitoring
 - Secure remote access
 - Portfolio hosting
 - Real-time system monitoring APIs
 - Learning platform for self-hosting and software engineering
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Owner

Name: Ramesh

Role: Software Engineer (.NET / WPF / ASP.NET)

Experience: 5+ Years

Primary Objectives:

1. Self-host services on low-power hardware
 2. Automate Tamil movie acquisition and management
 3. Learn DevOps and infrastructure concepts
 4. Build portfolio projects using real production-like data
 5. Create a central platform for experimentation and automation
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Hardware

Primary Server

Raspberry Pi 3B

Purpose:

- Main homelab server
- Runs 24/7

Responsibilities:

- Hosts all self-hosted services
 - Runs media automation stack
 - Provides monitoring APIs
 - Serves portfolio integrations
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Operating System

CasaOS

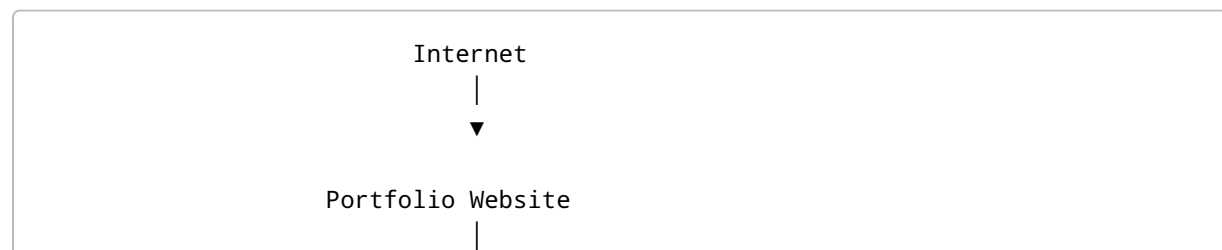
Purpose:

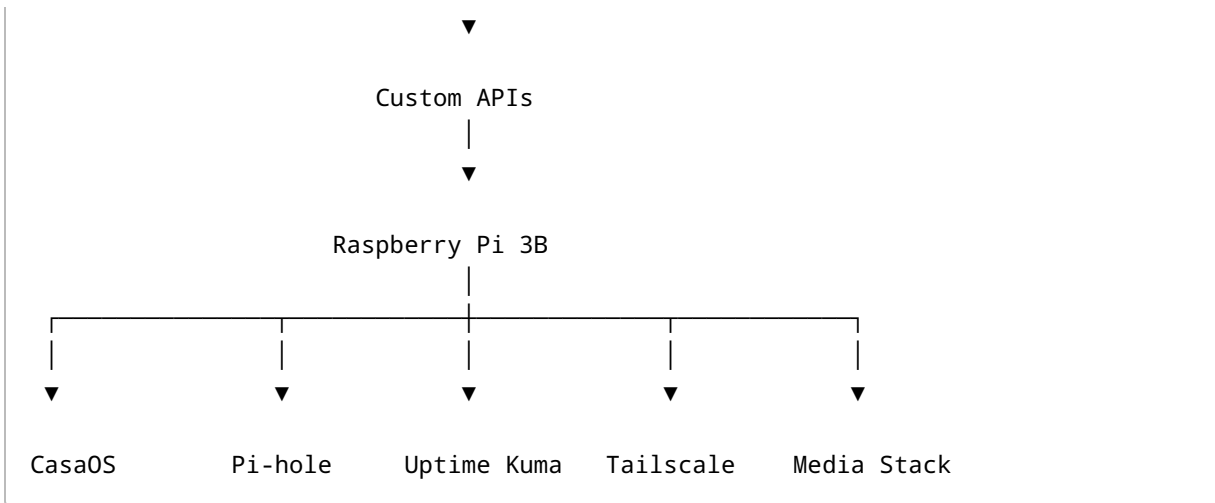
Container and application management platform.

Benefits:

- Easy service deployment
 - Web-based management
 - Docker container support
 - Centralized application dashboard
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High-Level Architecture





Core Infrastructure Services

1. Tailscale

Purpose:

Secure remote access to the homelab.

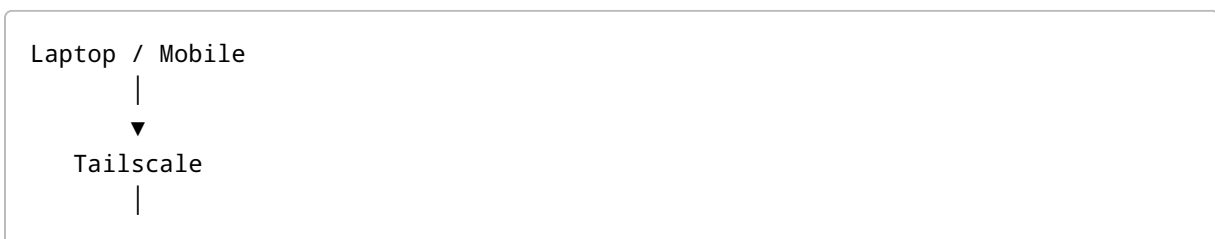
Capabilities:

- Access Raspberry Pi remotely
- Access CasaOS dashboard
- Access Emby
- Access Pi-hole
- Access monitoring dashboards

Benefits:

- No port forwarding
- Encrypted communication
- Private VPN network

Usage:



2. Pi-hole

Purpose:

Network-wide ad and tracker blocking.

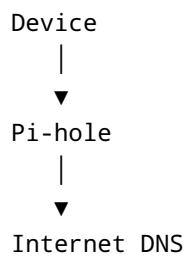
Functions:

- DNS sinkhole
- Ad blocking
- Tracker blocking
- DNS analytics

Benefits:

- Faster browsing
- Cleaner web experience
- Reduced tracking

Flow:



3. Uptime Kuma

Purpose:

Monitor all hosted services.

Monitored Targets:

- Raspberry Pi
- CasaOS

- Emby
- Radarr
- Jackett
- Pi-hole
- Portfolio APIs
- Website endpoints

Features:

- Uptime monitoring
- Downtime alerts
- Service status dashboard
- Historical availability

Media Automation Stack

Goal:

Automated Tamil movie discovery with manual approval before downloading.

1. Radarr

Purpose:

Movie management and automation.

Responsibilities:

- Track wanted movies
- Search releases
- Manage metadata
- Coordinate downloads

Root Folder:

`/mnt/media/movies/tamil`

2. Jackett

Purpose:

Torrent indexer aggregation.

Responsibilities:

- Connect to multiple torrent sites
- Provide Torznab feeds to Radarr

Connection Type:

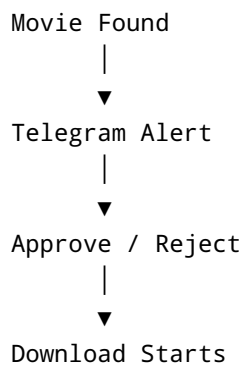
Torznab

3. Telegram Bot

Purpose:

Manual approval workflow.

Process:



Benefits:

- Avoid fake torrents
 - Avoid CAM releases
 - Quality control
 - Language verification
-

4. Download Client

Transmission

Primary lightweight torrent client.

Benefits:

- Raspberry Pi friendly
- Low memory usage
- Reliable

Default Port:

9091

Alternative Discussed:

- qBittorrent
-

5. Emby

Purpose:

Media server and streaming platform.

Library Path:

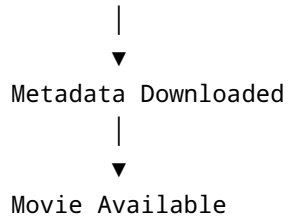
/mnt/media/movies/tamil

Features:

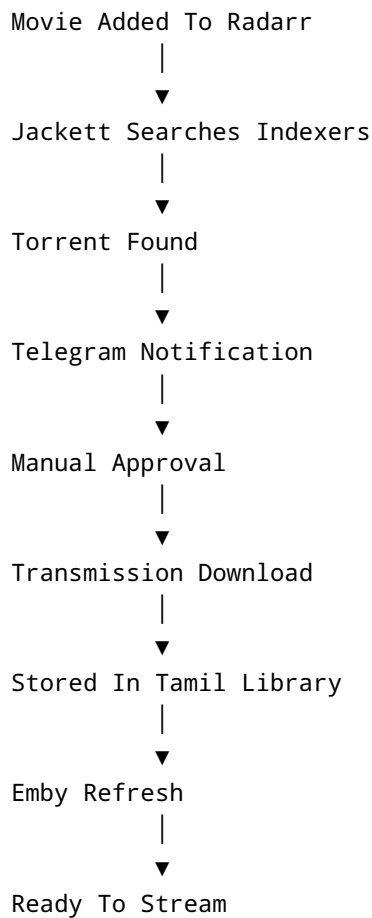
- Metadata management
- Automatic library refresh
- Streaming across devices

Flow:

```
Download Complete
|
▼
Emby Detects File
```



Media Workflow



Storage Layout

```
/mnt/media
└─ movies
```



```
└─ tamil
    ├── Movie 1
    ├── Movie 2
    └─ Movie 3
```

Used By:

- Radarr
- Transmission
- Emby

Portfolio Website

Purpose

Professional portfolio website showcasing:

- Experience
- Projects
- Technical skills
- Live server statistics

Live System Dashboard

The portfolio site consumes APIs hosted by the Raspberry Pi.

Displayed Metrics May Include:

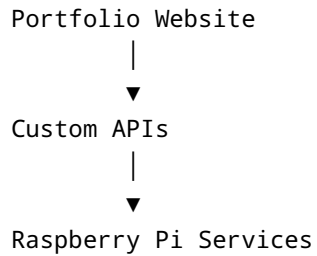
- Server uptime
- CPU usage
- RAM usage
- Storage utilization
- Active services
- Media library statistics
- Health status

Custom Monitoring APIs

Purpose:

Expose real-time homelab information.

Architecture:



Example Endpoints:

```
/api/system  
/api/cpu  
/api/memory  
/api/storage  
/api/uptime  
/api/services
```

Example Response:

```
{  
  "cpu": 32,  
  "memory": 58,  
  "disk": 71,  
  "uptime": "14 days"  
}
```

Security Model

Internal Access

Protected through:

- Tailscale

Public Access

Limited to:

- Portfolio Website
- Selected APIs

Security Goals:

- Avoid exposing infrastructure
 - Encrypt remote access
 - Minimize attack surface
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Monitoring Strategy

Uptime Kuma

Monitors:

- Infrastructure
- Applications
- APIs

Portfolio Dashboard

Displays:

- Real-time system metrics

Pi-hole

Monitors:

- DNS activity
 - Ad blocking statistics
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Design Principles

Low-Power Computing

Hardware:

- Raspberry Pi 3B

Goal:

Maximum functionality with minimal power consumption.

Self-Hosting

Preference for:

- Local ownership
 - Open-source software
 - Privacy-first solutions
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Automation With Human Approval

Downloads should not start automatically.

Reason:

- Quality assurance
 - Language verification
 - Torrent validation
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Learning Platform

Homelab is used to gain experience in:

- Linux
- Docker
- CasaOS
- Networking
- DNS
- Monitoring
- APIs

- Automation
 - System Administration
-

Future Enhancements

Planned

Sonarr

Tamil TV show automation.

Overseerr / Jellyseerr

Media request portal.

Telegram Inline Controls

Approve downloads directly from chat.

Automatic Torrent Cleanup

Remove completed torrents after seeding targets.

Advanced Metrics

Expose more system statistics through APIs.

Additional Services

Potential future self-hosted applications running under CasaOS.

Complete Technology Stack

Hardware

Raspberry Pi 3B

Platform

CasaOS

Remote Access

Tailscale

DNS Filtering

Pi-hole

Monitoring

Uptime Kuma

Media Server

Emby

Movie Automation

Radarr

Indexer Aggregation

Jackett

Download Client

Transmission

Notifications

Telegram Bot

Custom Development

Monitoring APIs

Public Presence

Portfolio Website

Media Type

Tamil Movies

Storage

/mnt/media/movies/tamil

Summary

This homelab is a multi-purpose self-hosted environment that combines media automation, infrastructure monitoring, network services, secure remote access, and custom software development on a Raspberry Pi 3B.

It serves as both a production system for daily use and a practical learning environment for software engineering, DevOps, networking, and self-hosting technologies.